

1. General Description

The EM74AHC1G240 and EM74AHCT1G240 are 1-bit inverting buffer/line driver with 3-state output. The device features an output enable $\overline{OE}$. A HIGH on $\overline{OE}$ causes the output to assume a high-impedance OFF-state. Inputs are overvoltage tolerant. This feature allows the use of these devices as translators in mixed voltage environments.

The AHC device has CMOS input switching levels and supply voltage range 2 V to 5.5 V.

The AHCT device has TTL input switching levels and supply voltage range 4.5 V to 5.5 V.

2. Features and Benefits

- Wide supply voltage range from 2.0 V to 5.5 V
- Overvoltage tolerant inputs to 5.5 V
- High noise immunity
- CMOS low power dissipation
- Latch-up performance exceeds 200 mA
- Symmetrical output impedance
- Balanced propagation delays
- Input levels:
 - For EM74AHC1G240: CMOS level
 - For EM74AHCT1G240: TTL level
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 3A exceeds 6000 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options

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Single inverting buffer/line driver; 3-state

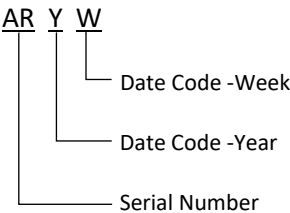
3. Ordering Information

Table 1. Ordering information

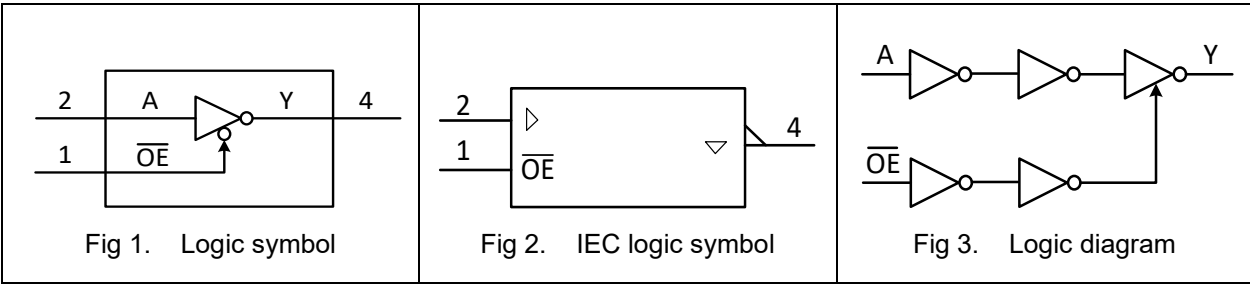
Type number	Topside marking	Package		
		Name	Description	Quantity
EM74AHC1G240GV	ARYW	SOT23-5L	SOT23 package, 5 pins 2.92 mm × 1.6 mm; 1.25 mm (Max) height	3000
EM74AHCT1G240GV	CRYW			
EM74AHC1G240GW	ARYW	SOT353	SOT353 package, 5 pins 2.1 mm × 1.25 mm; 1.1 mm (Max) height	3000
EM74AHCT1G240GW	CRYW			

MARKING INFORMATION

NOTE: YW = Date Code.

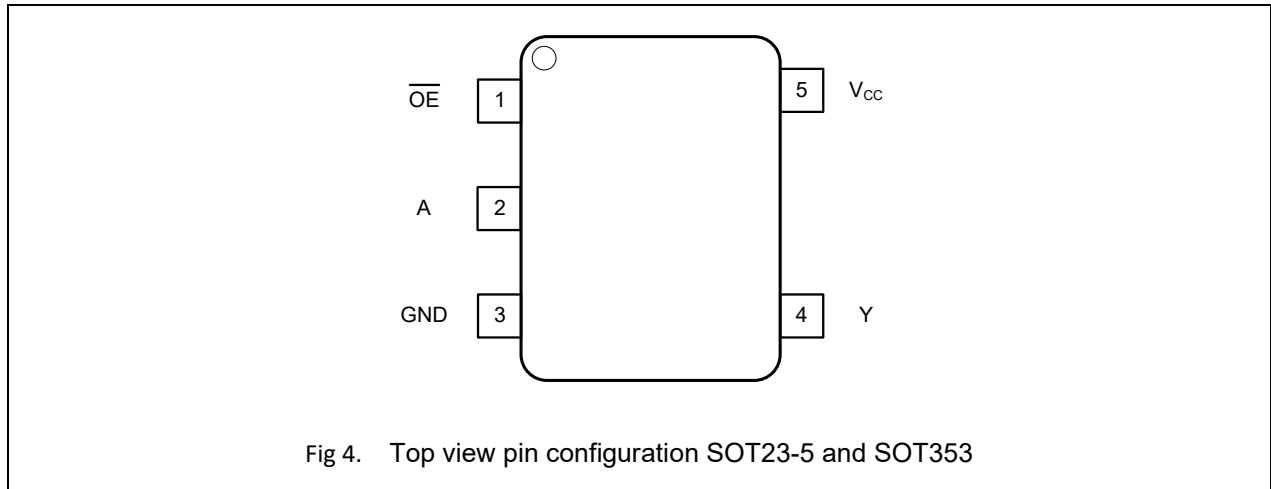


4. Function Diagram



5. Pinning Information

5.1. Pin map



5.2. Pin description

Table 2. Pin description

Symbol	Pin	
	SOT23-5, SOT353	Description
$\overline{OE}$	1	Output enable input
A	2	Data input
GND	3	Ground (0V)
Y	4	Data output
V _{CC}	5	Supply voltage

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; X = don't care; Z = high-impedance OFF-state.

Input		Output
$\overline{OE}$	A	Y
L	L	H
L	H	L
H	X	Z

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7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	7.0	V
V_I	input voltage		-0.5	7.0	V
I_{IK}	input clamping current	$V_I < -0.5\text{ V}$	-20		mA
I_{OK}	output clamping current	$V_O < -0.5\text{ V}$ or $V_O > V_{CC} + 0.5\text{ V}$ [1]		± 20	mA
I_O	output current	$-0.5\text{ V} < V_O < V_{CC} + 0.5\text{ V}$		± 25	mA
I_{CC}	supply current			75	mA
I_{GND}	ground current		-75		mA
P_{tot}	total power dissipation	$T_{amb} = -40\text{ }^{\circ}\text{C}$ to $+125\text{ }^{\circ}\text{C}$		250	mW
T_{stg}	storage temperature		-65	150	$^{\circ}\text{C}$

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	EM74AHC1G240			EM74AHCT1G240			Unit
			Min	Typ	Max	Min	Typ	Max	
V_{CC}	supply voltage		2.0	5.0	5.5	2.0	5.0	5.5	V
V_I	input voltage		0		5.5	0		5.5	V
V_O	output voltage		0		V_{CC}	0		V_{CC}	V
T_{amb}	ambient temperature		-40	25	125	-40	25	125	$^{\circ}\text{C}$
$\Delta t/\Delta V$	input transition rise and fall rate	$V_{CC} = 3.3\text{ V} \pm 0.3\text{ V}$			100			100	ns/V
		$V_{CC} = 5.0\text{ V} \pm 0.5\text{ V}$			20			20	ns/V

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9.Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74AHC1G240								
V _{IH}	HIGH-level input voltage	V _{CC} = 2.0 V	1.5			1.5		V
		V _{CC} = 3.0 V	2.1			2.1		V
		V _{CC} = 5.5 V	3.85			3.85		V
V _{IL}	LOW-level input voltage	V _{CC} = 2.0 V			0.5		0.5	V
		V _{CC} = 3.0 V			0.9		0.9	V
		V _{CC} = 5.5 V			1.65		1.65	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = -50 µA; V _{CC} = 2.0 V	1.9	2.0		1.9		V
		I _O = -50 µA; V _{CC} = 3.0 V	2.9	3.0		2.9		V
		I _O = -50 µA; V _{CC} = 4.5 V	4.4	4.5		4.4		V
		I _O = -4.0 mA; V _{CC} = 3.0 V	2.48	2.93		2.40		V
		I _O = -8.0 mA; V _{CC} = 4.5 V	3.80	4.39		3.70		V
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = 50 µA; V _{CC} = 2.0 V		0	0.1		0.1	V
		I _O = 50 µA; V _{CC} = 3.0 V		0	0.1		0.1	V
		I _O = 50 µA; V _{CC} = 4.5 V		0	0.1		0.1	V
		I _O = 4.0 mA; V _{CC} = 3.0 V		0.05	0.44		0.55	V
		I _O = 8.0 mA; V _{CC} = 4.5 V		0.07	0.44		0.55	V
I _{OZ}	OFF-state output current	V _{CC} = 5.5 V; V _I = V _{CC} or GND		0.01	2.5		10	µA
I _I	input leakage current	V _I = 5.5 V or GND ; V _{CC} = 0 V to 5.5 V		±0.01	±1.0		±2.0	µA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0 A ; V _{CC} = 5.5 V		0.01	10		40	µA
C _I	input capacitance			3.5				pF

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74AHCT1G240								
V _{IH}	HIGH-level input voltage	V _{CC} = 4.5 V to 5.5 V	2.0			2.0		V
V _{IL}	LOW-level input voltage	V _{CC} = 4.5 V to 5.5 V			0.8		0.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL} ;						
		I _O = -50 µA; V _{CC} = 4.5 V	4.4	4.5		4.4		V
		I _O = -8.0 mA; V _{CC} = 4.5 V	3.80	4.39		3.70		V
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL} ;						
		I _O = 50 µA; V _{CC} = 4.5 V		0	0.1		0.1	V
		I _O = 8.0 mA; V _{CC} = 4.5 V		0	0.44		0.55	V
I _{OZ}	OFF-state output current	V _{CC} = 5.5 V; V _I = V _{CC} or GND		0.01	2.5		10	µA
I _I	input leakage current	V _I = 5.5 V or GND ; V _{CC} = 0 V to 5.5 V		±0.01	±1.0		±2.0	µA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0 A ; V _{CC} = 5.5 V		0.01	10		40	µA
ΔI _{CC}	additional supply current	per input pin ; V _I = 3.4 V; other inputs at V _{CC} or GND; I _O = 0 A; V _{CC} = 5.5 V		0.23	1.5		1.5	mA
C _I	input capacitance			3.5				pF

[1]All typical values are measured at $T_{amb} = 25^\circ\text{C}$.

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10. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 7.

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74AHC1G240								
t _{pd}	propagation delay	A to Y; see Fig. 5 [2]						
		V _{CC} = 3.0 V to 3.6 V, C _L = 15 pF	1.0	5.5	8.5	1.0	9.0	ns
		V _{CC} = 4.5 V to 5.5 V, C _L = 15 pF	1.0	3.8	5.5	1.0	6.0	ns
t _{en}	enable time	$\overline{OE}$ to Y; see Fig. 6 [3]						
		V _{CC} = 3.0 V to 3.6 V, C _L = 15 pF	1.0	6.1	10.0	1.0	11.0	ns
		V _{CC} = 4.5 V to 5.5 V, C _L = 15 pF	1.0	4.6	6.5	1.0	7.0	ns
t _{dis}	disable time	$\overline{OE}$ to Y; see Fig. 6 [4]						
		V _{CC} = 3.0 V to 3.6 V, C _L = 15 pF	1.0	7.8	14.0	1.0	14.5	ns
		V _{CC} = 4.5 V to 5.5 V, C _L = 15 pF	1.0	4.9	7.5	1.0	8.0	ns
C _{PD}	power dissipation capacitance	C _L = 15 pF ; f = 1MHz ; V _I = GND to V _{CC} ; [5]		20				pF
EM74AHCT1G240								
t _{pd}	propagation delay	A to Y; see Fig. 5 [2]						
		V _{CC} = 4.5 V to 5.5 V, C _L = 15 pF	1.0	5.4	7.5	1.0	8.0	ns
t _{en}	enable time	$\overline{OE}$ to Y; see Fig. 6 [3]						
		V _{CC} = 4.5 V to 5.5 V, C _L = 15 pF	4.0	8.0	11	4.0	11.5	ns
t _{dis}	disable time	$\overline{OE}$ to Y; see Fig. 6 [4]						
		V _{CC} = 4.5 V to 5.5 V, C _L = 15 pF	2.0	4.1	6.5	2.0	7.0	ns
C _{PD}	power dissipation capacitance	C _L = 15 pF ; f = 1MHz ; V _I = GND to V _{CC} ; [5]		21				pF

[1] Typical values are measured at $T_{amb} = 25\text{ °C}$ and $V_{CC} = 3.3\text{ V}$ and 5.0 V respectively.

[2] t_{pd} is the same as t_{PLH} and t_{PHL} .

[3] t_{en} is the same as t_{PZH} and t_{PZL} .

[4] t_{dis} is the same as t_{PLZ} and t_{PHZ} .

[5] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \Sigma(C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

C_L = output load capacitance in pF;

V_{CC} = supply voltage in V;

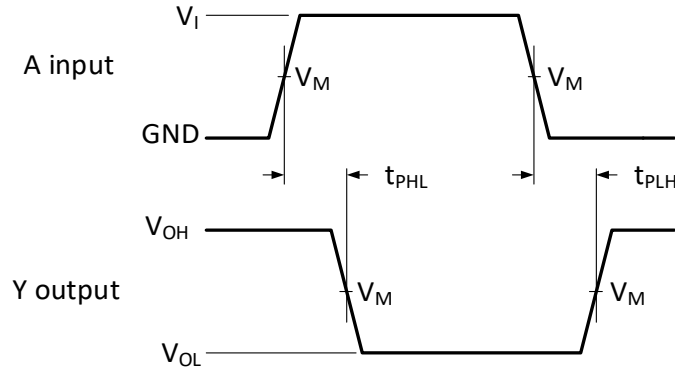
N = number of inputs switching;

$\Sigma(C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

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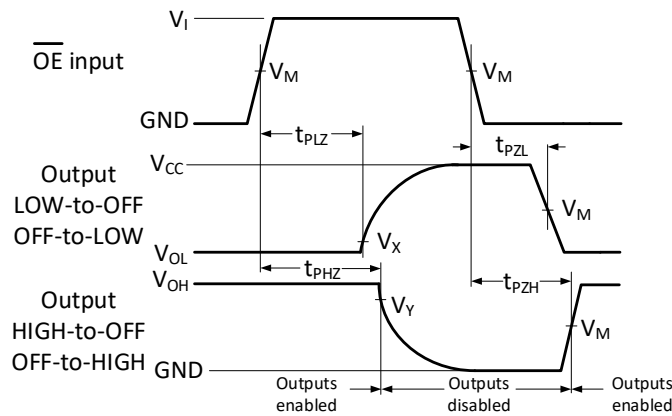
10.1. Waveforms and test circuit



Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig 5. The input A to output Y propagation delays



Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

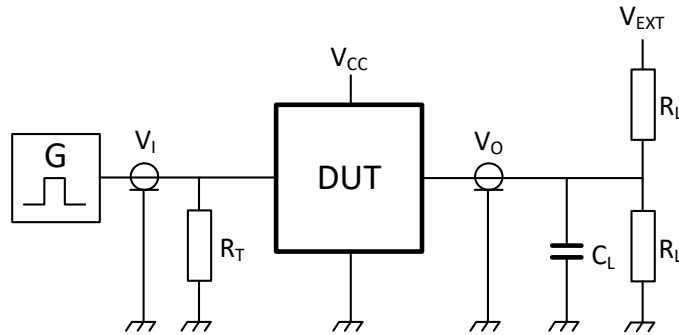
Fig 6. 3-state enable and disable times

Table 8. Measurement points

Type	Input		Output		
	V_I	V_M	V_M	V_X	V_Y
EM74AHC1G240	GND to V_{CC}	$0.5 \times V_{CC}$	$0.5 \times V_{CC}$	$V_{OL} + 0.3 \text{ V}$	$V_{OH} - 0.3 \text{ V}$
EM74AHCT1G240	GND to 3.0 V	1.5 V	$0.5 \times V_{CC}$	$V_{OL} + 0.3 \text{ V}$	$V_{OH} - 0.3 \text{ V}$

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Test data is given in Table 9.

Definitions for test circuit:

R_L = Load resistance.

C_L = Load capacitance including jig and probe capacitance.

R_T = Termination resistance should be equal to the output impedance Z_o of the pulse generator.

V_{EXT} = External voltage for measuring switching times.

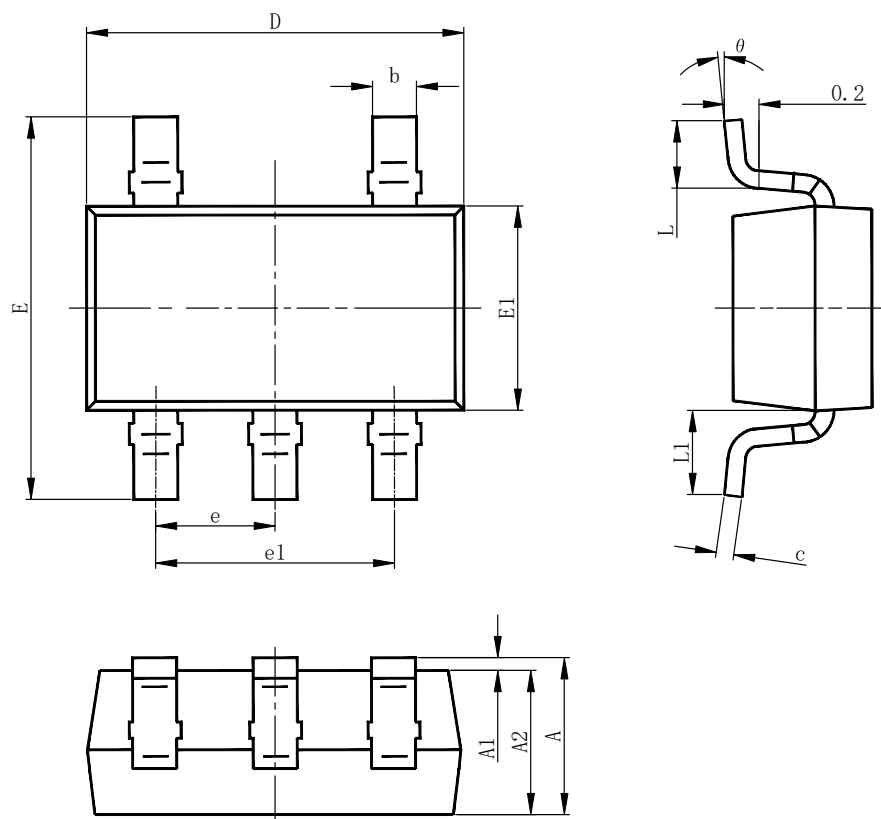
Fig 7. Test circuit for measuring switching times

Table 9. Test data

Type	Input	Load		V_{EXT}		
	$t_r = t_f$	C_L	R_L	t_{PLH}, t_{PHL}	t_{PZH}, t_{PHZ}	t_{PZL}, t_{PLZ}
EM74AHC1G240	≤ 2.5 ns	15 pF	500 Ω	open	GND	$2V_{CC}$
EM74AHCT1G240	≤ 2.5 ns	15 pF	500 Ω	open	GND	$2V_{CC}$

11. Package Outline

SOT23-5L

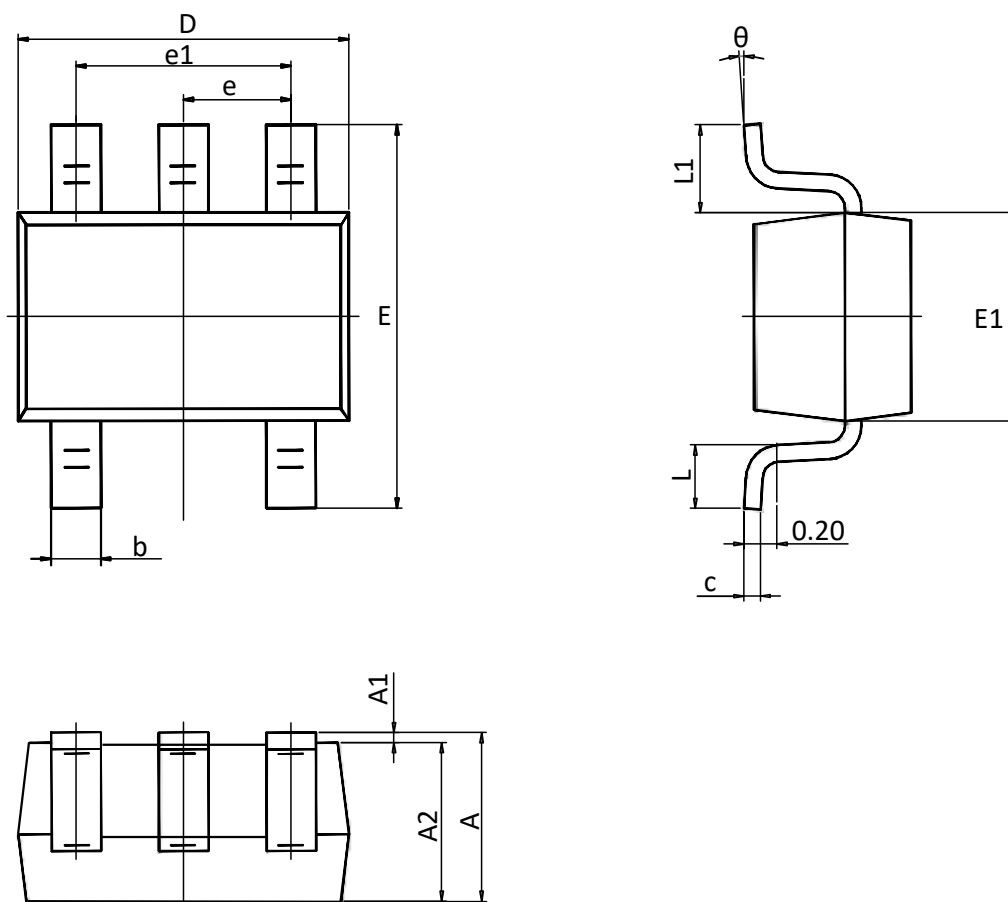


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E1	1.500	1.700	0.059	0.067
E	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
L1	0.600REF.		0.024REF.	
θ	0°	8°	0°	8°

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SOT353



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	0.900	1.100	0.035	0.043
A1	0.000	0.100	0.000	0.004
A2	0.900	1.000	0.035	0.039
b	0.150	0.350	0.006	0.014
c	0.110	0.175	0.004	0.007
D	2.000	2.200	0.079	0.087
E	2.150	2.450	0.085	0.096
E1	1.150	1.350	0.045	0.053
e	0.650 TYP.		0.026 TYP.	
e1	1.200	1.400	0.047	0.055
L	0.260	0.460	0.010	0.018
L1	0.525 REF.		0.021 REF.	
theta	0°	8°	0°	8°

Tape and Reel Information

12.1. Carrier tape dimensions

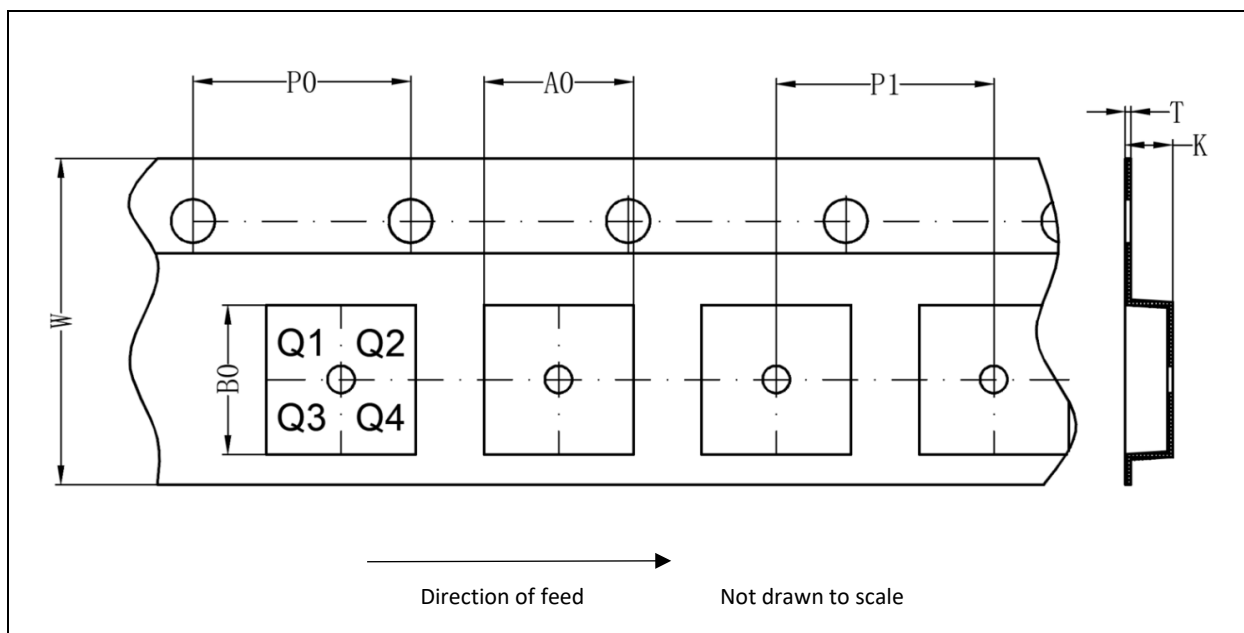


Table 10. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOT23-5L	3.23	3.17	1.37	0.25	4	8	4	Q3
SOT353	2.55	2.55	1.2	0.2	4	8	4	Q3

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12.2. Reel and box dimensions

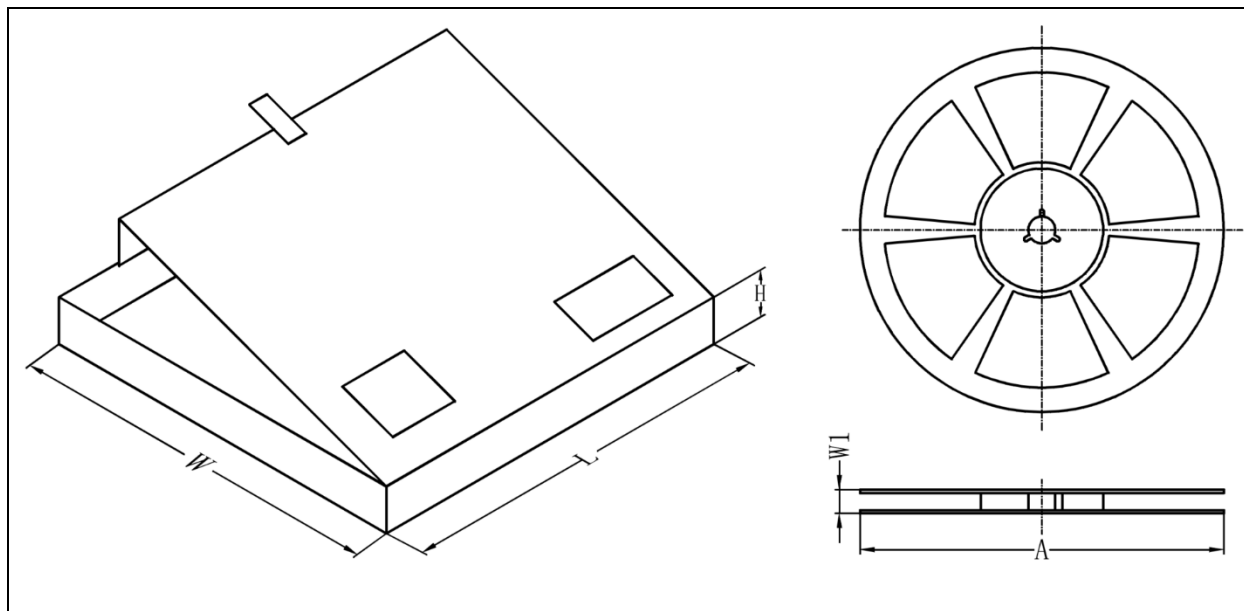


Table11. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Max Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm)[2]
SOT23-5L	GV	180	13.2	3000	1	210x200x40
SOT353	GW	180	13.2	3000	1	210x200x40

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

EM74AHC1G240; EM74AHCT1G240

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13. Abbreviations

Table 12. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	Electro Static Discharge
HBM	Human Body Model
MM	Machine Model
TTL	Transistor-Transistor Logic

14. Revision History

Table 13. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74AHC_AHCT1G240 Rev 1.0	Jun 15, 2024	Product datasheet		